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§12. Chinese remainder theorem, maximal ideals, prime ideals, principal ideal domains

Chinese remainder theorem

Recall special case. if $n_1,\cdots,n_k$ are pairwise coprime positive integers,

then $\mathbb{Z}\to\mathbb{Z}/n_1\mathbb{Z}\times\cdots\times\mathbb{Z}/n_k\mathbb{Z}$ is surjective homom. of rings

and the kernel is $n_1\mathbb{Z}\times\cdots\times n_k\mathbb{Z}=n_1\cdots n_k\mathbb{Z}$

DEF Let R be a commutative ring.

Two ideals I,J of R are called comaximal (coprime) if $I+J=R$.

ie. $1\in R$ can be written as $1=a+b$, $a\in J$, $b\in J$.

eg. $R=\mathbb{Z}$. $I=(m)=m\mathbb{Z}$, $J=(n)=n\mathbb{Z}$, st. $\gcd(m,n)=1\Leftrightarrow(m)+(n)=(m,n)=(1)\Leftrightarrow 1=mx+ny$ for $x,y\in\mathbb{Z}$.

THM (Chinese remainder theorem). Let $I_1,\cdots,I_k$ be ideals of a commutative ring R.

Then the natural map $\Phi: R\to R/I_1\times\cdots\times R/I_k$, $x\mapsto(x \bmod I_1,\cdots,x \bmod I_k)$ is a homom. of rings with $\ker\Phi=I_1\cap\cdots\cap I_k$. 到此为止是显然的

Further if $I_1,\cdots,I_k$ are pairwise comaximal then

1) Φ is surjective

2) $I_1\cap\cdots\cap I_k=I_1\cdots I_k$

$\Rightarrow\overline{\Phi}: R/\underbrace{I_1\cap\cdots\cap I_k}_{\text{kernel}}=R/I_1\cdots I_k\simeq R/I_1\times\cdots\times R/I_k$.

Pf. First assume $k=2$. $I_1I_2\subset I_1\cap I_2$ ✓

$I_1\cap I_2\subsetneq I_1I_2$ since $R=I_1+I_2\Rightarrow 1=a_1+a_2$, $a_1\in I_1, a_2\in I_2$. $\forall b\in I_1\cap I_2$, $b=\underset{\in I_1I_2}{ba_1}+\underset{\in I_1I_2}{ba_2}\in I_1I_2$

Now consider $\Phi(a_1)=(a_1 \bmod I_1, \underset{\|}{a_1} \bmod I_2)=(0,1)$, with $a_1\mapsto a_2$, $(0,1)\in R/I_1\times R/I_2$

$\Phi(a_2)=(\underset{\|}{a_2} \bmod I_1, a_2 \bmod I_2)=(1,0)$, with $a_2\mapsto a_1$, $(1,0)\in R/I_1\times R/I_2$

$\Rightarrow$ any $(x_1 \bmod I_1, x_2 \bmod I_2)=\Phi(a_2x_1+a_1x_2)$.

In general, we induction on k.

$\Phi: R\xrightarrow{?}R/I_1\times R/I_2\cdots I_k\xrightarrow[\text{hypothesis}]{\text{induction}}R/I_1\times\cdots\times R/I_k$. $\ker\Phi=I_1\cap I_2\cdots I_k\supsetneq I_1\cdots I_k$.

Check: $I_1+I_2\cdots I_k=R$.

Write $1=a_i+b_i$ for $a_i\in I_1$, $b_i\in I_i$, $i=2,\cdots,k$.

$\Rightarrow 1=(a_2+b_2)\cdots(a_k+b_k)=\underbrace{a_2\cdots a_k+\text{terms with } a_i}_{\in I_1}+\underbrace{b_2\cdots b_k}_{\in I_2\cdots I_k}$ $\square$

Some logic.

DEF A partial order on a nonempty set A is a relation $\leq$ on A, st. $\forall x,y,z\in A$,

1) $x\leq x$ (reflexive)

2) if $x\leq y$ and $y\leq x$, then $x=y$ (antisymmetric)

3) if $x\leq y$ and $y\leq z$, then $x\leq z$ (transitive)

A chain (totally ordered subset) is a subset B of A where $\forall x,y\in B$, either $x\leq y$ or $y\leq x$.

Zorn's lemma ($\Leftrightarrow$ Axiom of choice)

if A is a partially ordered set ("poset") in which every chain B has an upper bound (ie. $\exists a\in A$, st. $b\leq a$, $\forall b\in B$)

then A has a maximal element x (ie. $\nexists y\in A$, $x\leq y$ and $x\neq y$).

Maximal ideals.

DEF Let R be a ring. A two-sided ideal m of R is called maximal if $m\subsetneq R$ and the only two-sided ideals containing m are m and R.

PROP Every proper two-sided ideal $I\subsetneq R$ is contained in a maximal ideal.

Pf Let $S:=\{\text{proper two-sided ideals of } R \text{ containing } I\}$ poset by inclusion.

Every chain $\{J_i\}$ in S has an upper bound $J:=\bigcup J_i\in S$

Zorn's lemma $\Rightarrow S$ admits a maximal element $m\Rightarrow m$ is a maximal ideal containing I. $\square$

PROP Suppose that R is commutative,

Then an ideal $m\subset R$ is maximal $\Leftrightarrow R/m$ is a field.

Pf $m\subset R$ is maximal $\Leftrightarrow R/m=:\overline{R}$ has only two ideals (0), $(1)=\overline{R}$.

$\Leftrightarrow \overline{R}$ is a field

"$\Leftarrow$": obvious.

"$\Rightarrow$": $\forall a\in\overline{R}$, $a\neq 0$, we have $(a)\neq(0)\Rightarrow(a)=(1)$ ie. $\exists a'\in\overline{R}$, st. $aa'=1\Rightarrow a\in R^\times\Rightarrow\overline{R}$ is a field. $\square$

REM if R is noncommutative, R/m is a division ring $\Rightarrow$ m maximal

(eg. $R=Mat_n(\mathbb{C})$ has only two two-sided ideals $\{0\}, R$)

eg. 1) $R=\mathbb{Z}$. p prime number. $(p)=p\mathbb{Z}$ is a maximal ideal.

2) $R=\mathbb{Z}[x]$. $(p)=p\mathbb{Z}[x]$ is not maximal

$\subsetneq (p,x)$ or $(p,f(x))$

maximal — any polynomial which is irreducible mod p

3) $R=\mathbb{R}[x]$. (x^2+1) is maximal. $\mathbb{R}[x]/(x^2+1)\simeq\mathbb{C}$. (impose $x^2=-1$)

4) G finite group. $R=\mathbb{C}[G]\supset I_R=(g-1)_{g\in G}$ is a maximal (two-sided) ideal. $R/I_R\simeq\mathbb{C}$. (I_R: augmentation ideal)

Prime ideals. Assume from now on that R is commutative.

DEF A proper ideal $p\subsetneq R$ is called a prime ideal if

$\forall a,b\in R$, $ab\in p\Rightarrow a\in p$ or $b\in p$.

eg. p prime number. $(p)=p\mathbb{Z}$ is a prime ideal of $\mathbb{Z}$. ($p|ab\Rightarrow p|a$ or $p|b$).

(0) is also a prime ideal of $\mathbb{Z}$.

PROP An ideal $p\subset R$ is prime $\Leftrightarrow$ R/p is an integral domain. (注意我们约定整环中 $0\neq 1$, 故"1不为素数".)

Pf. $\pi: R\to R/p$, $a\mapsto\bar{a}$. $a,b\in R$ with $ab\in p\Leftrightarrow \bar{a}\bar{b}=0$.

$a\in p$ or $b\in p\Leftrightarrow \bar{a}=0$ or $\bar{b}=0$ □

COR A maximal ideal is always prime.

Properties of prime ideals

PROP 1) (prime avoidance lemma)

Let $p_1,\cdots,p_n$ be finitely many prime ideals and let I be an ideal of R, st. $I\subset\bigcup_{i=1}^n p_i$

Then $I\subset p_i$ for some i.

2) Let $I_1,\cdots,I_n$ be finitely many ideals and let p be a prime ideal of R, st. $p\supset\bigcap_{i=1}^n I_i$

Then $p\supset I_i$ for some i.

Pf. 2) suppose not. $\forall i, \exists x_i\in I_i$, $x_i\notin p$

But $x_1 \cdots x_n \in I_1 \cdots I_n \subset \bigcap_{i=1}^n I_i \subset \mathfrak{p}$, contradicting $\mathfrak{p}$ prime.

1). We prove by induction on n that $I \not\subset \mathfrak{p}_i$ for $i=1,\cdots,n \Rightarrow I \not\subset \bigcup_{i=1}^n \mathfrak{p}_i$.

$n=1$ ✓. Suppose $n-1$ ✓.

inductive hypothesis $\Rightarrow \forall i, \exists x_i \in I$ but $x_i \notin \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_{i-1} \cup \mathfrak{p}_{i+1} \cup \cdots \cup \mathfrak{p}_n$.

if some $x_i \notin \mathfrak{p}_i$, we are done. So assume $x_i \in \mathfrak{p}_i, \forall i$.

Now consider $y := x_1 \cdots x_{n-1} + x_n \in I$ $\quad x_1 \cdots x_{n-1} \notin \mathfrak{p}_n$ since $\mathfrak{p}_n$ prime. $x_n \in \mathfrak{p}_n \Rightarrow y \notin \mathfrak{p}_n$.

for $1 \le i \le n-1$, $x_1 \cdots x_{n-1} \in \mathfrak{p}_i$, $x_n \notin \mathfrak{p}_i \Rightarrow y \notin \mathfrak{p}_i$. $\quad$ so $y \notin \mathfrak{p}_i$ for any i. $\square$

PROP. Let $\Phi: R \to S$ be a homom. of commutative rings.

(Recall: $I \subset S$ ideal $\Rightarrow \Phi^{-1}(I) \subset R$ ideal)

If $\mathfrak{p} \subset S$ is a prime ideal. Then $\Phi^{-1}(\mathfrak{p}) \subset R$ is prime.

Pf: $R/\Phi^{-1}(\mathfrak{p}) \hookrightarrow S/\mathfrak{p}$ (integral domain since $\mathfrak{p}$ prime.)

先看 $R \xrightarrow{\Phi} S \to S/\mathfrak{p}$ (composite Ψ). $\ker \Psi = \Phi^{-1}(\mathfrak{p})$

$\Rightarrow R/\Phi^{-1}(\mathfrak{p})$ is also an integral domain $\Rightarrow \Phi^{-1}(\mathfrak{p})$ prime. $\square$

Zariski topology

DEF. Let R be a commutative ring.

We set 齊 $\mathrm{Spec}(R) := \{\text{prime ideals } \mathfrak{p} \subset R\}$.

We define a topology on $\mathrm{Spec}(R)$ as follow:

- The closed subsets are of the form $V(I) := \{\mathfrak{p} \in \mathrm{Spec}(R) \mid \mathfrak{p} \supset I\}$ for some ideal $I \subset R$. (nonnecessarily prime)
- Equivalently, the open subsets are of the form $D(I) := \{\mathfrak{p} \in \mathrm{Spec}(R) \mid \mathfrak{p} \not\supset I\} = V(I)^c$ for some ideal $I \subset R$.

LEM/DEF. 1) $V(\{0\}) = \mathrm{Spec}(R)$ $\quad V(R) = \emptyset$.

2) Let $\{I_j\}_{j \in S}$ be ideals of R

$\bigcap_{j \in S} V(I_j) = V\left(\sum_{j \in S} I_j\right)$ where $\sum_{j \in S} I_j := \{\text{finite sums } \sum_{j \in S} a_j,\ a_j \in I_j\}$.

3) Let $I_1, \cdots, I_n$ be finitely many ideals of R

Then $\bigcup_{i=1}^{n} V(I_i) = V(\bigcap_{i=1}^{n} I_i)$.

in particular, the $V(I)$ form the closed subsets of a topology on $\mathrm{Spec}(R)$ called the Zariski topology.

Pf. 1) obvious.

2) observe that $I \subset J \Rightarrow V(I) \supset V(J)$.

So $V(I_j) \supset V(\sum_{j\in S} I_j) \Rightarrow \bigcap_{j\in S} V(I_j) \supset V(\sum_{j\in S} I_j)$

for "$\subset$" $\forall p \in \bigcap_{j\in S} V(I_j)$ we have $p \supset I_j$, $\forall j \in S$. $\Rightarrow p \supset \sum_{j\in S} I_j \Rightarrow p \in V(\sum_{j\in S} I_j)$.

3) Similarly, $V(I_i) \subset V(\bigcap_{i=1}^{n} I_i) \Rightarrow \bigcup_{i=1}^{n} V(I_i) \subset V(\bigcap_{i=1}^{n} I_i)$

for "$\supset$" $\forall p \in V(\bigcap_{i=1}^{n} I_i)$, we have $p \supset \bigcap_{i=1}^{n} I_i$ previous PROP $\Rightarrow p \supset I_i$ for some $i \Rightarrow p \in \bigcup_{i=1}^{n} V(I_i)$. □

REM Let $\Phi: R \to S$ be a homom. of commutative rings.

Then Φ induces $f: \mathrm{Spec}(S) \to \mathrm{Spec}(R)$, $p \mapsto \Phi^{-1}(p)$

which is a continuous map for the Zariski topology.